Relational Charge on the Spectral Semiring: Gauge Rigidity, Coherence Types, and a Reference-Free Parity Floor

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Abstract

Which part of an algebraic integer's phase structure belongs to the number itself, and which only to a chosen reference ray? We refactor the emission algebra's charge into a purely relational invariant — the pair charge $t_{\mathrm{rel}}(\alpha, \beta) = t(\alpha) - t(\beta) \in \mathbb{Q}/\mathbb{Z}$, with $t(\alpha) = \frac{1}{2\pi} \operatorname{Arg} \alpha$, defined when rational — and determine exactly how much survives. The relational data is a groupoid cocycle whose trivializations are the absolute charges; the reference ray is a gauge choice. Rigidity: conjugation-closed spectra force every internally rational object to be absolutely admissible, with anchor $n \in \{m, 2m\}$ over the relational group $\mathbb{Z}/m\mathbb{Z}$ — the absolute theory is the canonical section — and the composition laws descend verbatim. On the inadmissible sector the theory is genuinely finer: objects carry a computable coherence type, and a *modulus-pinning theorem* — an irreducible polynomial with a uniquely attained root modulus admits no root-of-unity ratio between distinct roots — proves every Salem number relationally inert and no two Salem numbers circle-locked, while irreducible non-inert inadmissible objects do exist (twisted shells $q(x^k)$).

All of this is executed exactly: a shell-complete, float-free contact procedure corroborates the pinning theorem on forty-six certified instances, including an exhaustive degree-12 small-coefficient census (37/37 inert), cross-checked in a second computer-algebra stack. Last — and conditionally on the companion note's emission gap, the one hypothesis not proved here beyond degree two — the parity-graded Mahler floor is evaluated reference-free: the parity bit is the golden pair's internal order-2 relation $t_{\mathrm{rel}}(\varphi, \varphi') = \frac{1}{2}$, a sign twist absorbs the second anchor sector, and $\mu_{\mathrm{rel}} = \mu$; that identity and the attainments are unconditional, only the numerical floor values rest on the gap. Unconditional proofs, gap-conditional results, and per-instance computations are tagged apart throughout.

1 Introduction

1.1 The question

The emission algebra of the companion note [1] grades its objects by two orthogonal readouts: a magnitude (the Mahler measure M, subject to the emission gap $\mathrm{M} \in \{1\} \cup [\varphi, \infty)$ with $\varphi = \frac{1+\sqrt{5}}{2}$) and a phase (a charge χ valued in a finite cyclic group $\mathbb{Z}/n\mathbb{Z}$, defined when every conjugate's argument lies on a finite rational-angle lattice). Both are *object-attached*: the charge of a conjugate α is read against the fixed reference ray $\operatorname{Arg} = 0$, and an object either possesses a charge group or — as for every Salem number [4, 8] — possesses none.

This note asks what changes when the phase character is attached instead to the *relationship between oscillators*. Each conjugate $\alpha = |\alpha|\, e^{2\pi i\, t(\alpha)}$ is an oscillator with growth $|\alpha|$ and phase

$t(\alpha) \in \mathbb{R}/\mathbb{Z}$; the relational datum is the phase *difference*

$$t_{\mathrm{rel}}(\alpha, \beta) \;=\; t(\alpha) - t(\beta) \;\in\; \mathbb{Q}/\mathbb{Z},$$

declared whenever the difference is rational, with no reference ray anywhere. Three questions organize the study:

(i) **Gauge.** In what precise sense is the absolute charge a gauge-fixing of the relational one, and what plays the role of the gauge group?

(ii) **Rigidity.** Does the relational theory admit objects the absolute theory rejects — oscillator systems internally in tune but globally out of tune with the reference ray?

(iii) **Content.** Which theorems of the absolute theory — the lcm composition law, the Adams/CRT projectors, the parity-graded floor $\mu(n) \in \{\varphi, 2\}$ — are secretly relational, and what genuinely new invariants appear?

1.2 Contributions

- **The gauge structure** (§3). The relational charge is a $\mathbb{Q}/\mathbb{Z}$-valued cocycle on a coherence groupoid; absolute charge is a trivialization; the reference ray is the gauge; a global rotation $\alpha \mapsto e^{2\pi i c}\alpha$ shifts every absolute phase by c and fixes every relative one. What the language buys is the exact content of §4: rigidity says the gauge is not free on the algebraic category — an object's phases are pinned to rational values outright, the only structural slack being the anchor dichotomy $n \in \{m, 2m\}$, itself absorbed object-by-object by the sign twist (Lemma 6.3). [forced]

- **Rigidity** (§4). Integrality fixes the gauge: for a conjugation-closed spectrum, all pairwise differences rational forces all phases rational. Relational admissibility therefore coincides with absolute admissibility, with the absolute group pinned to an anchor $n \in \{m, 2m\}$ over the relational group $\mathbb{Z}/m\mathbb{Z}$; both anchors occur, and the relational group can be a proper subgroup of the absolute one (witness: $x^4 + 5x^2 + 5$, absolute $\mathbb{Z}/4\mathbb{Z}$, relational $\mathbb{Z}/2\mathbb{Z}$, $\mathrm{M} = 5$). Read this as a canonicity theorem, not a redundancy: it proves the absolute framework of [1] is the unique natural presentation of the relational one on its own domain, while the relational language retains purchase exactly where absolute methods have none — the inadmissible sector. [forced]

- **Descent of the composition laws** (§5). Coherence classes are preserved by the Adams operations, which multiply relative charges by k; the tensor law $\Delta(A \otimes B) \supseteq \Delta(A) + \Delta(B)$ holds with equality on admissible inputs (the lcm law, relationally), while strict growth occurs exactly through *offset cancellation* — the tensor product internalizes cross-object phase relations. [forced]

- **A reference-free parity floor** (§6). The parity bit of the charge–measure coupling is intrinsic: it is the realized difference $t_{\mathrm{rel}}(\varphi, \varphi') = \frac{1}{2}$ (since $\varphi/\varphi' = -\varphi^2$), not the lattice membership of the ray π. A sign-twist involution $p(x) \mapsto (-1)^{\deg p} p(-x)$ exchanges the two anchor sectors at odd m and preserves the Mahler measure, so the relational floor equals the absolute one: $\mu_{\mathrm{rel}}(m) = \varphi$ for even m (attained), and $\mu_{\mathrm{rel}}(m) = \mu(m)$ for odd m. Identity and attainment [forced]; the floor *values* are conditional per Remark 2.3

- **Coherence types of inadmissible objects** (§7). On the sector the absolute theory discards, the relational theory is genuinely finer: every object carries a finite coherence type, computable exactly on equal-modulus shells by a ratio-resultant / cyclotomic-contact procedure with a provable completeness bound; and a *modulus-pinning theorem* — irreducibility plus a uniquely attained root modulus forbids torsion ratios between distinct roots — proves *every* Salem number relationally inert [forced], corroborated on forty-six certified instances (degrees 4–10 including β_4 and Lehmer's polynomial, an exhaustive degree-12 small-coefficient census of 37, and five larger-coefficient spot checks). Irreducible non-inert inadmissible objects exist: twisted shells $q(x^k)$, witness $x^4 + x^2 + 2$.

- **Cross-object locking** (§8). Two objects can be phase-locked while each is internally inert; the notion is decidable on circle shells by a mixed ratio resultant. The pinning theorem's mixed form proves no two distinct Salem numbers circle-lock [forced]; the six pairwise scans among the canonical instances stand as its exact corroboration.

1.3 Relation to the companion note

This note is a refactoring and an extension of [1], not a replacement. Every statement of [1] about admissible objects is recovered here, usually in a sharper, reference-free form; the new content lives (a) in the gauge/rigidity analysis, which explains *why* the absolute theory loses nothing on its own domain, and (b) on the inadmissible sector, where the absolute theory is silent by construction and the relational theory supports nontrivial, decidable invariants. Notation follows [1] throughout. Because [1] is a technical note of the same project — DOI-archived but not peer-reviewed — this note is organized so that its verification burden is explicit, and the conditional content it supports is confined to exactly two numerical values (Remark 6.6). The unconditional results (§§3–5 and §§7–8) are fully auditable within this document. The conditional results (the floor *values* in Theorem 6.5) are auditable here in the sense that their dependence on the named hypothesis (EG) is explicit and traceable (Remark 2.3, Appendix C); verifying (EG) itself beyond the quadratic case reproduced in Appendix D requires [1], archived at DOI `10.5281/zenodo.21121863` (within the project repository deposit, v1.0.0).

Relation to the wider literature. A rational phase difference between conjugates is, equivalently, a torsion multiplicative relation $\alpha_i = \zeta\, \alpha_j$ between conjugates. Multiplicative relations connecting conjugate algebraic numbers are a classical subject (Smyth [11] and the literature following it, notably the torsion-quotient program of Dubickas [14, 15, 16]; Remark 7.19 records the attribution); the present note can be read as isolating their torsion part, packaging it as a groupoid-valued invariant, and supplying a shell-complete exact decision procedure for it. On the Salem question specifically, the shell-torsion half of inertness is exactly *non-degeneracy* in the sense of linear recurrence sequences [13]; Remark 7.19 records the bridge, the modulus-pinning proof, and what the coherence type adds beyond it. The census infrastructure connects to the small-Salem tables originating with Boyd [9] and extended by Mossinghoff [12]; the unconditional floor landscape below the gap is Dobrowolski's bound [10]. We are not aware of a prior treatment of the coherence type as an invariant of charge-inadmissible objects; the project's standing citation-provenance discipline applies (see the provenance block following the references).

1.4 A note on rigor

Every numerical assertion below was verified in exact arithmetic (§9); in this note the entire verification path is integer and rational polynomial arithmetic — no floating-point number appears anywhere, in decisions or in display. Each result carries an epistemic tag: [forced] (proved), [computed] (verified exhaustively over a stated finite range), [plausible] (strongly evidenced, not proved in full generality)[1], [open] (open). Statements inheriting a caveated status from [1] say so explicitly, and Appendix C lists every statement inherited from [1] with the status tagged there and the place it enters this note. The conditional content of this note is confined to the two floor evaluations of Theorem 6.5 (Remark 6.6); every structural statement — rigidity, gauge, descent, coherence types, locking — is unconditional. We do not claim to resolve Lehmer's problem or the general no-Salem statement; we characterize their interaction with the relational phase structure.

2 Preliminaries

2.1 Objects and operators

Fix the ring $\overline{\mathbb{Z}}$ of algebraic integers. An *object* is a finite multiset $O = \{\alpha_1, \ldots, \alpha_n\}$ of nonzero algebraic integers that is closed under complex conjugation — equivalently, the root multiset of a monic $p \in \mathbb{Z}[x]$ with $p(0) \neq 0$ (not necessarily irreducible), equivalently the spectrum of an integer matrix. The algebra's operations are those of [1]: direct sum $\oplus$ (multiset union; on polynomials, the product), tensor $\otimes$ (eigenvalue products; Kronecker product of matrices), and the Adams operations ψ^k ($\lambda \mapsto \lambda^k$ on each eigenvalue), $k \geq 1$. The semiring/λ-ring structure of these operations — associativity, distributivity of $\otimes$ over $\oplus$, identities, and $\psi^a \circ \psi^b = \psi^{ab}$ — is established in [1] and is not re-verified here; this note uses only the operations and their effect on angles, $t(\alpha\beta) = t(\alpha) + t(\beta)$ and $t(\lambda^k) = k\, t(\lambda)$, not the axiomatic package.

2.2 The angle coordinate and the absolute charge

Definition 2.1 (Angle)**.** For $\alpha \in \mathbb{C}^\times$ write $t(\alpha) = \frac{1}{2\pi} \operatorname{Arg} \alpha \in \mathbb{R}/\mathbb{Z}$, so $\alpha = |\alpha|\, e^{2\pi i\, t(\alpha)}$.

Because p has real coefficients, the angle multiset $t(O)$ is closed under negation in $\mathbb{R}/\mathbb{Z}$; we use this constantly.

Definition 2.2 (Absolute charge, [1, Def. 2.4])**.** An object is *charge-admissible* if every conjugate α satisfies $\alpha^n \in \mathbb{R}_{>0}$ for some $n \geq 1$, i.e. $t(\alpha) \in \mathbb{Q}/\mathbb{Z}$ for every conjugate. The *charge group* is $\mathbb{Z}/n\mathbb{Z}$ for n the least common denominator of the $t(\alpha)$, and the charge of α is $\chi_n(\alpha) = n\, t(\alpha) \bmod n$.

The magnitude character is the Mahler measure [5] $\mathrm{M}(O) = \prod_i \max(1, |\alpha_i|)$, with Kronecker's theorem ($\mathrm{M} = 1$ iff every root is a root of unity) and the emission gap $\mathrm{M} \in \{1\} \cup [\varphi, \infty)$ carrying the statuses recorded in [1, §2, §8]. We write $\mu(n)$ for the absolute floor of [1, Thm. 5.1]: the infimum of $\mathrm{M} > 1$ over charge-admissible objects with charge group $\mathbb{Z}/n\mathbb{Z}$.

Remark 2.3 (Status of the emission gap, and the classical floors)*.* Three floor statements are in play and must not be conflated. (i) Kronecker's characterization of $\mathrm{M} = 1$ is classical and proven [2]. (ii) The unconditional classical lower bound is Smyth's theorem [6]: a *non-reciprocal* monic integer polynomial has $\mathrm{M} \geq \theta_0 = 1.32471\ldots$, the real root of $x^3 - x - 1$ (the smallest Pisot number, often called the plastic number; we refer to θ_0 as Smyth's constant); for reciprocal polynomials the existence of any uniform bound > 1 is Lehmer's problem [3], open. (iii) The emission gap used here — $\mathrm{M} \in \{1\} \cup [\varphi, \infty)$ *for charge-admissible objects* — is [1, Prop. 2.3], treated here

[1] That is, [plausible] marks a conjecture with a specified evidence level: supported by exhaustive computation over stated ranges and by structural argument, but not proved. The tag system is project-internal; this footnote fixes its dictionary.

as a named hypothesis (EG) (its quadratic case is elementary and reproduced self-containedly in Appendix D): [forced] for the integer-quadratic spectrum, [computed]/[plausible] in general. Since $\varphi > \theta_0$, statement (iii) is not implied by Smyth's theorem even on the non-reciprocal part; nor is it Lehmer's conjecture, being restricted to the rational-angle sector. In interval terms: Smyth's theorem already excludes $(1, \theta_0)$ for every non-reciprocal polynomial; (EG) additionally excludes $[\theta_0, \varphi)$ on the charge-admissible sector, where it binds reciprocal and non-reciprocal objects alike. Consequently, wherever this note evaluates a floor *as* φ, the attainment is [forced] while the identification of the infimum as φ (rather than some value in $[\theta_0, \varphi)$) is conditional on (EG); Theorem 6.5 and the table of §10 carry this tag explicitly.

3 The relational charge

3.1 Coherence, classes, and the relational group

Definition 3.1 (Pair charge and coherence)**.** Conjugates $\alpha, \beta \in O$ are *coherent*, written $\alpha \approx \beta$, if $t(\alpha) - t(\beta) \in \mathbb{Q}/\mathbb{Z}$; in that case their *relative charge* is

$$t_{\mathrm{rel}}(\alpha, \beta) \;=\; t(\alpha) - t(\beta) \;\in\; \mathbb{Q}/\mathbb{Z}.$$

Coherence is an equivalence relation ($t(\alpha) - t(\alpha) = 0$; symmetry by negation; transitivity because rationals are closed under addition), so O partitions into *coherence classes* $C_1, \ldots, C_r$.

Definition 3.2 (Class groups, relational group, coherence type)**.** For a class C, its *class group* is the subgroup $\Delta(C) = \langle\, t_{\mathrm{rel}}(\alpha, \beta) : \alpha, \beta \in C \,\rangle \le \mathbb{Q}/\mathbb{Z}$. The *relational charge group* of O is $\Delta(O) = \sum_i \Delta(C_i) \le \mathbb{Q}/\mathbb{Z}$. The object is *relationally admissible* if it has a single coherence class. The *coherence type* of O is the triple: the partition $\{C_i\}$, the class groups $\{\Delta(C_i)\}$, and the involution that negation induces on the classes (Theorem 4.3).

Example 3.3. For $x^3 - 2$ the angles are $\{0, \frac{1}{3}, \frac{2}{3}\}$: every pairwise difference is a multiple of $\frac{1}{3}$, so there is a single class with $\Delta = \mathbb{Z}/3\mathbb{Z}$, fixed by the negation involution. For $x^4 + 5x^2 + 5$ (Example 4.5) the angles are $\{\frac{1}{4}, \frac{3}{4}\}$: again a single class, now with $\Delta = \mathbb{Z}/2\mathbb{Z}$. For a Salem quartic such as β_4 the type is computed in §7: rational block $\{\tau, 1/\tau\}$ plus two mirrored circle singletons.

Proposition 3.4 (Cyclicity, relational form)**.** *Every class group and every relational group is a finite cyclic group $\mathbb{Z}/m\mathbb{Z}$; no non-cyclic group occurs.* [forced]

Proof. Each is a subgroup of $\mathbb{Q}/\mathbb{Z}$ generated by finitely many rationals, hence a finite subgroup of $\mathbb{Q}/\mathbb{Z}$; $\mathbb{Q}/\mathbb{Z}$ has exactly one subgroup of each finite order, namely $\langle 1/m \rangle \cong \mathbb{Z}/m\mathbb{Z}$. This is [1, Thm. 3.1] with the generators taken to be differences rather than absolute angles. □

3.2 The cocycle, the trivialization, and the gauge

Proposition 3.5 (Cocycle and trivialization)**.**

(i) On coherent triples, $t_{\mathrm{rel}}(\alpha, \beta) + t_{\mathrm{rel}}(\beta, \gamma) = t_{\mathrm{rel}}(\alpha, \gamma)$: the relative charge is a $\mathbb{Q}/\mathbb{Z}$-valued cocycle on the coherence groupoid (objects: the conjugates; morphisms: coherent pairs).

(ii) If O is charge-admissible with charge group $\mathbb{Z}/n\mathbb{Z}$, then on every pair $t_{\mathrm{rel}}(\alpha, \beta) = \frac{1}{n}\big(\chi_n(\alpha) - \chi_n(\beta)\big)$: the relational charge is the coboundary of the absolute one.

(iii) Any two functions $s : C \to \mathbb{Q}/\mathbb{Z}$ with $s(\alpha) - s(\beta) = t_{\mathrm{rel}}(\alpha, \beta)$ on a class C differ by a constant.

(iv) The rotation $\rho_c : \alpha \mapsto e^{2\pi i c}\alpha$ (an operation on the ambient analytic picture; for irrational c it leaves the category of algebraic integers) shifts every absolute angle by c and fixes every relative charge. [forced]

Proof. (i) and (iii) are immediate from Definition 3.1; (ii) is Definition 2.2; (iv) is $t(\rho_c\alpha) = t(\alpha) + c$. ☐

We call the choice of the reference ray $\mathrm{Arg} = 0$ — equivalently, the choice of constant in (iii) — the *gauge*. The absolute theory of [1] is the relational theory plus this gauge choice. The terminology is structural, not physical: the circle acts as a structure group on phase assignments, Proposition 3.5(iii) exhibits the trivializations as a torsor under it, and integrality (§4) provides the canonical section; no connection to physical gauge theory is implied. A reader uncomfortable with the vocabulary may substitute "differences of normalized arguments" throughout without loss. The cocycle language earns its keep twice over: it is the functorial carrier for the descent analysis of §5, and it reduces the proof of Theorem 4.1 to a single coset computation. The next section shows that on the algebra's own objects the choice is almost costless: integrality fixes the gauge.

4 Rigidity: integrality fixes the gauge

Theorem 4.1 (Rigidity)**.** *Let O be an object (conjugation-closed) that is relationally admissible with relational group $\Delta(O) = \mathbb{Z}/m\mathbb{Z}$. Then every angle $t(\alpha)$ is rational: O is charge-admissible, and its absolute charge group $\mathbb{Z}/n\mathbb{Z}$ satisfies*

$$n \in \{m,\ 2m\}.$$

Both anchors occur. [forced]

Proof. All pairwise differences lie in $\frac{1}{m}\mathbb{Z}/\mathbb{Z}$. Fix any conjugate α. Since O is closed under complex conjugation, $\beta_0 := \overline{\alpha}$ is among the conjugates, with $t(\beta_0) = -t(\alpha)$; since O is relationally admissible — a single coherence class — every pair of conjugates is coherent, in particular $\alpha \approx \beta_0$, so $t(\alpha) - (-t(\alpha)) = 2\,t(\alpha) \in \frac{1}{m}\mathbb{Z}/\mathbb{Z}$, whence $t(\alpha) \in \frac{1}{2m}\mathbb{Z}/\mathbb{Z}$. Thus all angles are rational and lie in $\frac{1}{2m}\mathbb{Z}/\mathbb{Z}$, so the least common denominator n divides $2m$. And m divides n: every angle lies in $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$, hence every difference does; the differences generate $\frac{1}{m}\mathbb{Z}/\mathbb{Z}$, so $\frac{1}{m}\mathbb{Z}/\mathbb{Z} \subseteq \frac{1}{n}\mathbb{Z}/\mathbb{Z}$, i.e. $m \mid n$. Hence $n \in \{m, 2m\}$. For the anchors: $x^m - 2$ has angles $\{j/m\}$, so $n = m = \Delta$; $x^3 + 2$ has angles $\{\frac{1}{6}, \frac{1}{2}, \frac{5}{6}\}$, so $n = 6$ while the differences generate $\frac{1}{3}\mathbb{Z}/\mathbb{Z}$, i.e. $m = 3$, $n = 2m$ (ledger entry B). ☐

Corollary 4.2. *Relational admissibility coincides with absolute admissibility on the objects of the algebra. In particular the relational theory admits no new objects: an oscillator system that is internally in tune (all pairwise phase differences rational) is automatically in tune with the reference ray.* [forced]

The mechanism deserves emphasis: it is the *integrality* of the object — real coefficients, hence conjugation-closed spectra — that pins the gauge. The rotations ρ_c of Proposition 3.5(iv) act freely on the analytic envelope, but the algebraic locus meets each orbit only at rational offsets. Rigidity is thus a feature, not a limitation: it is the uniqueness half of a gauge-fixing statement —

the absolute theory of [1] is the canonical section — and it locates the relational theory's genuine domain, the inadmissible sector of §7, where there is no section to fix. Equivalently, rigidity is a *completeness guarantee*: no information is lost in the relational reformulation — the prerequisite for trusting it on the sector where it extends reach.

Theorem 4.3 (Structure of partial coherence)**.** *Negation $t \mapsto -t$ permutes the coherence classes of any object. A class C with $-C = C$ has all angles rational ($t(C) \subset \mathbb{Q}/\mathbb{Z}$); a class with $-C \neq C$ carries an offset: $t(C) \subseteq c + \frac{1}{m_C}\mathbb{Z}/\mathbb{Z}$ for a class constant c that is irrational (well defined modulo $\frac{1}{m_C}\mathbb{Z}/\mathbb{Z}$), and $t(-C) = -t(C)$ carries the offset $-c$. Every object therefore decomposes into a* rational block *(the union of the self-paired classes, which is charge-admissible on its own) and mirrored pairs of irrationally offset classes.* [forced]

Proof. Negation preserves rationality of differences, hence maps classes to classes. If $-C = C$ then for $\alpha \in C$ the angle $-t(\alpha)$ belongs to $t(C)$, so $2t(\alpha) \in \Delta(C) \subset \mathbb{Q}/\mathbb{Z}$ and $t(\alpha)$ is rational; conversely all differences inside a class being rational, one rational member makes all members rational. If $-C \neq C$, pick $\alpha_0 \in C$ and set $c = t(\alpha_0)$; every member differs from α_0 by an element of $\frac{1}{m_C}\mathbb{Z}/\mathbb{Z}$, giving the coset form, and c must be irrational or else C would be rational and self-paired by the previous sentence. □

Lemma 4.4 (Odd anchors are full)**.** *If O is charge-admissible with odd absolute group $\mathbb{Z}/n\mathbb{Z}$, then $\Delta(O) = \mathbb{Z}/n\mathbb{Z}$: at odd order the relational and absolute groups coincide.* [forced]

Proof. Say $\Delta(O) = \mathbb{Z}/d\mathbb{Z}$ with $d \mid n$. All angles lie in a single coset $t_0 + \frac{1}{d}\mathbb{Z}/\mathbb{Z}$ (differences generate $\frac{1}{d}\mathbb{Z}/\mathbb{Z}$ and the object is one class since all angles are rational). Negation closure forces $2t_0 \in \frac{1}{d}\mathbb{Z}/\mathbb{Z}$, so $t_0 \in \frac{1}{2d}\mathbb{Z}/\mathbb{Z}$ and the angles lie in $\frac{1}{2d}\mathbb{Z}/\mathbb{Z}$; hence $n \mid 2d$. With n odd this gives $n \mid d$, so $d = n$. □

Example 4.5 (The group drop, and its absence at odd order)**.** $g = x^4 + 5x^2 + 5$ is irreducible (Eisenstein at 5); its x^2-roots $\frac{-5\pm\sqrt{5}}{2}$ are both negative, so all four roots are purely imaginary: angles $\{\frac{1}{4}, \frac{3}{4}\}$, absolute group $\mathbb{Z}/4\mathbb{Z}$, differences $\{0, \frac{1}{2}\}$, relational group $\mathbb{Z}/2\mathbb{Z}$, and $\mathrm{M}(g) = 5$ (the squared moduli are $\frac{5\mp\sqrt{5}}{2} \approx 1.38$ and 3.62, both exceeding 1 exactly because $5 - \sqrt{5} > 2 \iff \sqrt{5} < 3$; hence $\mathrm{M} = |g(0)| = 5$). The absolute n thus conflates two invariants — the relational group and the anchor — which Lemma 4.4 shows can differ only at even n. (Ledger entry E; compare the catalog seed $K = x^4 + 5x^2 - 5$ of [1], whose sign-mirror this is: K keeps the full $\Delta = \mathbb{Z}/4\mathbb{Z}$ because its real pair sits at angles $\{0, \frac{1}{2}\}$.) [forced]

Corollary 4.6 (Salem geometry, relational form)**.** *Let τ be a Salem number of degree $2s+2$, with conjugates τ, $1/\tau$, and s unit-circle pairs $\lambda_j, \overline{\lambda_j}$. Then:*

(i) every circle conjugate has irrational angle (a unimodular number with rational angle is a root of unity, contradicting [1, Lem. 6.1]);

(ii) no circle conjugate is coherent with its own complex conjugate (their difference is $2t(\lambda_j)$, irrational), so circle classes come in mirrored pairs;

(iii) the rational block is exactly $\{\tau, 1/\tau\}$, with trivial class group (both angles are 0);

(iv) the only undetermined coherence is within *a half-block $\{\lambda_1, \dots, \lambda_s\}$ of circle conjugates — whether some $t(\lambda_i) - t(\lambda_j)$ is rational while both angles are irrational.*

[forced] *for (i)–(iii); (iv) is the question §7 decides per instance.*

5 Descent of the composition laws

Proposition 5.1 (Adams operations)**.** *For $k \geq 1$, the map ψ^k preserves the coherence partition (as a bijection of index sets) and multiplies every relative charge by k: $t_{\mathrm{rel}}(\alpha^k, \beta^k) = k\, t_{\mathrm{rel}}(\alpha, \beta)$, so $\Delta(\psi^k O) = k\,\Delta(O)$. In particular the CRT/Adams projectors of [1, Thm. 3.3] descend: for $\Delta(O) = \mathbb{Z}/m\mathbb{Z}$ with $m = \prod p_i^{e_i}$, the operator $\psi^{m/p_i^{e_i}}$ projects the relational group onto its p_i-primary component.* [forced]

Proof. $t(\alpha^k) - t(\beta^k) = k\big(t(\alpha) - t(\beta)\big)$ in $\mathbb{R}/\mathbb{Z}$, and for an integer $k \geq 1$ this is rational if and only if $t(\alpha) - t(\beta)$ is rational; so coherence is preserved and reflected, and the class map is the identity on indices. The group identity and the CRT statement follow as in [1, Thm. 3.3], applied to differences. □

Proposition 5.2 (Tensor: containment, the lcm law, and internalization)**.** *For objects A, B:*

(i) a pair $\big(\alpha\beta,\ \alpha'\beta'\big)$ in $A\otimes B$ is coherent iff $\big(t(\alpha)-t(\alpha')\big)+\big(t(\beta)-t(\beta')\big) \in \mathbb{Q}/\mathbb{Z}$; consequently $\Delta(A \otimes B) \supseteq \Delta(A) + \Delta(B)$.

(ii) If A and B are relationally admissible with groups $\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/k\mathbb{Z}$, then $A \otimes B$ is relationally admissible with $\Delta(A \otimes B) = \frac{1}{m}\mathbb{Z} + \frac{1}{k}\mathbb{Z} = \frac{1}{\mathrm{lcm}(m,k)}\mathbb{Z}$ modulo $\mathbb{Z}$: the lcm *law of [1, Thm. 4.2] is a relational statement, and safe composition $\Leftrightarrow$ coprime orders descends with it (*safe*: the two factor charges remain independently recoverable from the composite, [1, Thm. 4.3]).*

(iii) Strict containment in (i) occurs exactly through offset cancellation*: classes $C_A \subseteq A$, $C_B \subseteq B$ with irrational offsets c_A, c_B (Theorem 4.3) whose sum c_A+c_B is rational make all of $C_A \cdot C_B$ mutually coherent. Canonically, for any object O the square $O \otimes O$ contains, for each mirrored pair $(C, -C)$, the rational block $C \cdot (-C)$ — in particular the diagonal products $\alpha\overline{\alpha} = |\alpha|^2$ at angle 0. The tensor product internalizes cross-object phase relations, even between relationally inert factors.* [forced]

Proof. (i) is $t(\alpha\beta) = t(\alpha) + t(\beta)$; the containment takes $\beta = \beta'$ or $\alpha = \alpha'$. (ii): by rigidity (Theorem 4.1) admissible inputs have all angles rational, so every sum in (i) is rational and $A \otimes B$ is one class. For the group: fixing $\beta = \beta'$ realizes every element of $\Delta(A) = \frac{1}{m}\mathbb{Z}/\mathbb{Z}$ as a difference in $A \otimes B$, and fixing $\alpha = \alpha'$ realizes $\Delta(B) = \frac{1}{k}\mathbb{Z}/\mathbb{Z}$; conversely, by (i) every difference is a sum of one element of each. Hence $\Delta(A \otimes B) = \frac{1}{m}\mathbb{Z} + \frac{1}{k}\mathbb{Z}$ modulo $\mathbb{Z}$, which is $\frac{d}{mk}\mathbb{Z}/\mathbb{Z} = \frac{1}{\mathrm{lcm}(m,k)}\mathbb{Z}/\mathbb{Z}$ with $d = \gcd(m,k)$. Recoverability at coprime orders is the CRT projection, realized inside the algebra by the Adams operators (Proposition 5.1); at $\gcd(m,k) = d > 1$ the surjection $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z} \twoheadrightarrow \mathbb{Z}/\operatorname{lcm}(m,k)\mathbb{Z}$, given by $(a,b) \mapsto a\frac{L}{m} + b\frac{L}{k} \bmod L$ with $L = \mathrm{lcm}(m,k)$, has kernel cyclic of order d, generated by $(m/d,\ -k/d)$ (cf. [1, Thm. 4.3]). (iii): if $c_A + c_B \in \mathbb{Q}$ then any two products from $C_A \times C_B$ differ by an element of $\frac{1}{m_{C_A}}\mathbb{Z} + \frac{1}{m_{C_B}}\mathbb{Z} \subset \mathbb{Q}/\mathbb{Z}$; the mirrored pair of one object supplies $c + (-c) = 0$. □

Proposition 5.3 (Direct sum and phase locking)**.** *The coherence classes of $A \oplus B$ are generated by those of A and B together with the* cross edges *$\alpha \approx \beta$ ($\alpha \in A, \beta \in B$). If one cross pair between classes C_A, C_B is coherent then all of $C_A \times C_B$ is, and this happens iff the offsets satisfy $c_A - c_B \in \mathbb{Q}$ (rational blocks having offset 0). We call A and B* locked *along (C_A, C_B) in that case, and* circle-locked *when both classes consist of unimodular conjugates. Locking between rational blocks is automatic and carries no information; locking between irrationally offset classes is a genuine cross-object relational charge, invisible to the absolute theory on either factor.* [forced]

Proof. Transitivity through the class relations gives the all-or-nothing statement: $t(\alpha') - t(\beta') = \big(t(\alpha') - t(\alpha)\big) + \big(t(\alpha) - t(\beta)\big) + \big(t(\beta) - t(\beta')\big)$, a sum of rationals. The offset criterion restates the definition of the class constants. $\square$

Propositions 5.2(iii) and 5.3 are two views of one phenomenon: $\oplus$ records a lock as a merged class; $\otimes$ records it as a created rational block.

6 The parity floor, reference-free

Lemma 6.1 (The golden internal relation)**.** *The golden pair carries an intrinsic order-2 relational charge:*

$$\frac{\varphi}{\varphi'} = -\varphi^2, \qquad \textit{hence} \qquad t_{\mathrm{rel}}(\varphi, \varphi') = \tfrac{1}{2}.$$

[forced] *(ledger entry J)*

Proof. $\varphi' = -1/\varphi$, so $\varphi/\varphi' = -\varphi^2$, a negative real: relative angle π, relative charge $\frac{1}{2}$. $\square$

This is [1, Lem. 5.2] with the reference ray deleted: what the π-ray obstruction tracks is not the lattice membership of an absolute ray but the realized *difference* $\frac{1}{2}$ inside the golden pair. The floor-attaining construction of [1, Thm. 5.1] always realizes it:

Lemma 6.2 (q_k realizes the difference $\frac{1}{2}$)**.** *For every $k \geq 1$, $q_k = x^{2k} + x^k - 1$ has a real root of each sign (for odd k: $(1/\varphi)^{1/k} > 0$ and $-\varphi^{1/k} < 0$; for even k: $\pm(1/\varphi)^{1/k}$), hence realizes $t_{\mathrm{rel}} = \frac{1}{2}$; its angle set is all of $\frac{1}{2k}\mathbb{Z}/\mathbb{Z}$, so q_k is relationally admissible with $\Delta(q_k) = \mathbb{Z}/2k\mathbb{Z}$ (and anchor $n = 2k$).* [forced]

Proof. The x^k-substitution gives $y^2 + y - 1$ with roots $1/\varphi > 0$ and $-\varphi < 0$ (ledger entry K); the stated real roots follow by parity of k, and the full angle set is computed in [1, Thm. 5.1]: inert roots at angles j/k, measure-bearing roots at $(2j+1)/2k$. Differences include $\frac{1}{2k}$, so the class group is full. $\square$

Lemma 6.3 (The sign twist)**.** *The involution $T(p)(x) = (-1)^{\deg p}\, p(-x)$ on objects multiplies every root by -1; it therefore shifts every angle by $\frac{1}{2}$, fixes every difference (hence the coherence type and Δ), preserves the Mahler measure, and preserves monic integrality. For odd m it exchanges the two anchor sectors of Theorem 4.1: objects with $(\Delta, n) = (\mathbb{Z}/m\mathbb{Z},\ m)$ and those with $(\Delta, n) = (\mathbb{Z}/m\mathbb{Z},\ 2m)$ correspond bijectively.* [forced]

Proof. Only the anchor exchange needs an argument. An $n = 2m$ object at odd m has (by the proof of Theorem 4.1) angles in a coset $\frac{e}{2m} + \frac{1}{m}\mathbb{Z}/\mathbb{Z}$ with e odd. The shift by $\frac{1}{2} = \frac{m}{2m}$ moves the offset to $\frac{e+m}{2m}$; since e and m are both odd, $e + m$ is even, so $\frac{e+m}{2m} = \frac{(e+m)/2}{m} \in \frac{1}{m}\mathbb{Z}/\mathbb{Z}$: the image has $n = m$ (angles in $\frac{1}{m}\mathbb{Z}/\mathbb{Z}$ with differences generating it). Conversely the shift moves $\frac{1}{m}\mathbb{Z}/\mathbb{Z}$ into the odd coset. (The one-line arithmetic is nonetheless corroborated on the engine for $m \in \{3, 5, 7, 9\}$ — ledger entry L — per the project convention that every load-bearing step, however small, gets a machine check; $T(x^3 + 2) = x^3 - 2$ is ledger entry B.) $\square$

Definition 6.4. For $m \geq 1$ let $\mu_{\mathrm{rel}}(m) = \inf\big\{\mathrm{M}(O) \;:\; O \text{ relationally admissible},\ \Delta(O) \cong \mathbb{Z}/m\mathbb{Z},\ \mathrm{M}(O) > 1\big\}$.

Theorem 6.5 (Reference-free parity floor)**.**

$$\mu_{\mathrm{rel}}(m) = \begin{cases} \varphi, & m \text{ even (attained, by } q_{m/2}), \\ \mu(m), & m \text{ odd,} \end{cases}$$

where μ is the absolute floor of [1, Thm. 5.1]. Statuses: the even attainment is constructive [forced]*; the identification of the even infimum as φ (rather than a value in $[\theta_0, \varphi)$) is conditional on* (EG) *(Remark 2.3). For odd m the identity $\mu_{\mathrm{rel}}(m) = \mu(m)$ is* [forced]*, so the value inherits [1]: $\mu_{\mathrm{rel}}(3) = 2$* [forced]*, and $\mu_{\mathrm{rel}}(m) = 2$ for general odd m is* [computed] *(to the searched ranges of [1, §6]) /* [plausible]*.*

Proof. Even $m = 2k$. Lemma 6.2 gives an object with $\Delta = \mathbb{Z}/m\mathbb{Z}$ and $\mathrm{M} = \varphi$; rigidity makes every competitor charge-admissible, so the emission gap bounds $\mu_{\mathrm{rel}}(m) \geq \varphi$, with the status recorded in Remark 2.3.

Odd m. By Theorem 4.1 the domain splits into the anchor sectors $n = m$ and $n = 2m$. Lemma 4.4 identifies the first with the full family of charge-admissible objects of absolute group $\mathbb{Z}/m\mathbb{Z}$ — exactly the domain of $\mu(m)$ — and Lemma 6.3 maps the second onto the first bijectively and M-preservingly. Hence the infimum over the union equals $\mu(m)$. □

Remark 6.6 (Robustness if (EG) fails). The hypothesis (EG) enters exactly once: as the lower bound in the evaluation of the floors. If (EG) were to fail, every statement of §§3–5 and §§7–8 is unaffected, and within Theorem 6.5 the identity $\mu_{\mathrm{rel}}(\mathrm{odd}) = \mu(\mathrm{odd})$, the even attainment at φ (by q_k), and the odd attainment at 2 (by $x^m - 2$) all stand; only the numerical evaluations of the infima could move below the stated values. The conditional content of this note is confined to two numbers, not to any structure.

Corollary 6.7 (The parity bit is intrinsic)**.** *The charge–measure coupling of [1, Thm. 5.1, Cor. 6.3] is a relational statement: the universal floor φ is attainable at nontrivial relational charge iff the order-2 relation of the golden pair (Lemma 6.1) embeds in $\Delta(O)$, i.e. iff $\frac{1}{2} \in \Delta(O)$, i.e. iff $|\Delta(O)|$ is even. The single parity bit that governs the coupling is the order of a* relational *class, with no reference ray anywhere; and by the twist reduction the relational refinement adds no new burden to the companion's open odd-floor problem. Mechanism* [forced]*; universality inherits the caveats of [1, Thm. 5.1, Thm. 6.2].*

7 Coherence types on the inadmissible sector

By Corollary 4.2 the relational theory adds nothing on admissible objects. Its genuine domain is the sector the absolute theory discards — paradigmatically the Salem numbers, which by [1, Lem. 6.1] carry no absolute charge at all, yet by Definition 3.2 still carry a finite coherence type. This section makes the type computable and computes it for the two keystone occupants of the gap $(\varphi, 2)$.

7.1 Shells and the ratio object

Definition 7.1 (Shells)**.** The *shells* of O are the equivalence classes of conjugates under equal modulus. A ratio α/β is unimodular iff α, β share a shell.

Definition 7.2 (Ratio object)**.** For monic $p \in \mathbb{Z}[x]$ of degree n with $p(0) \neq 0$, the *ratio polynomial* is the primitive part of

$$\mathrm{Rat}_p(x) \;=\; \mathrm{Res}_y\big(p(y),\, p(xy)\big) \;\in\; \mathbb{Z}[x].$$

Because p is monic,

$$\mathrm{Res}_y\big(p(y),\, p(xy)\big) = \prod_i p(x\alpha_i) = \Big(\prod_i \alpha_i\Big)^n \prod_{i,j}\big(x - \alpha_j/\alpha_i\big), \qquad \prod_i \alpha_i = (-1)^n p(0) \neq 0,$$

so the degree is exactly n^2 (no drop: the leading coefficient is $(\pm p(0))^n$) and the root multiset is all n^2 ordered ratios α_j/α_i. When $p(0) = \pm 1$ the companion construction $\det\big(xI - C_p \otimes C_p^{-1}\big)$ (an integer matrix) has the same root multiset and serves as an independent second route (ledger entry G).

Standing assumption (this section): every p considered is squarefree — here each is irreducible. Lemma 7.5 uses this.

Lemma 7.3 (Contact criterion)**.** *Let α, β lie in a common shell. Then $t(\alpha) - t(\beta) \in \mathbb{Q}/\mathbb{Z} \iff \alpha/\beta$ is a root of unity $\iff$ some cyclotomic Φ_M with α/β among its roots divides* Rat_p*.* [forced]

Proof. α/β is unimodular; a unimodular complex number with rational angle a/b equals $e^{2\pi i a/b}$ and satisfies $z^b = 1$. Conversely roots of unity have rational angle. Divisibility: α/β is a root of Rat_p, and if it is a primitive M-th root of unity its minimal polynomial Φ_M divides Rat_p over $\mathbb{Q}$ (Gauss). □

Lemma 7.4 (Completeness of the contact scan)**.** *Let $P \in \mathbb{Z}[x]$ be nonzero. If $\Phi_M \mid P$ then $\phi(M) \leq \deg P$; and for every $M \geq 1$, $\phi(M) \geq \sqrt{M/2}$. Hence scanning all $M \leq 2(\deg P)^2$ decides, completely and in integer arithmetic, which cyclotomics divide P. For $P = \mathrm{Rat}_p$, of degree n^2, the bound is $2n^4$.* [forced]

Proof. The degree bound is $\deg \Phi_M = \phi(M) \leq \deg P$. For the totient bound — the claim, unambiguously, is $\phi(M) \geq \sqrt{M/2}$, i.e. $2\,\phi(M)^2 \geq M$ — write $M = 2^a M'$ with M' odd and use multiplicativity. On the 2-part, for $a \geq 1$, $\phi(2^a) = 2^{a-1}$ while $\sqrt{2^a/2} = 2^{(a-1)/2}$; since $a - 1 \geq (a-1)/2$ for $a \geq 1$ (equality only at $a = 1$), $\phi(2^a) \geq \sqrt{2^a/2}$. On each odd prime power, $\phi(p^b) = p^{b-1}(p-1) \geq p^{b-1}\sqrt{p} \geq p^{b/2} = \sqrt{p^b}$, using $p - 1 \geq \sqrt{p}$ for $p \geq 3$; multiplying over the odd part, $\phi(M') \geq \sqrt{M'}$. Assembling: if $a \geq 1$, $\phi(M) = \phi(2^a)\,\phi(M') \geq \sqrt{2^a/2}\cdot\sqrt{M'} = \sqrt{M/2}$; if $a = 0$, $\phi(M) \geq \sqrt{M} \geq \sqrt{M/2}$. The bound is tight at $M = 2$ ($\phi(2) = 1 = \sqrt{2/2}$), so the constant $\frac{1}{2}$ cannot be removed. Hence $\phi(M) \leq \deg P$ forces $M \leq 2(\deg P)^2$. (The inequality $2\phi(M)^2 \geq M$ is additionally corroborated by exhaustive integer sieve for all $M \leq 2\cdot 10^5$, comfortably past every scan bound used here, with $M = 2$ the unique tight case: ledger entry N.) □

Lemma 7.5 (Diagonal normalization)**.** *If p is squarefree, the multiplicity of $\Phi_1 = x - 1$ in* Rat_p *is exactly n (the diagonal ratios α_i/α_i). Any excess multiplicity, or any contact with Φ_M, $M \geq 2$, certifies a nontrivial torsion ratio.* [forced]

Proof. The diagonal ratios $\alpha_i/\alpha_i = 1$ account for n roots of $x - 1$; squarefreeness makes the α_i distinct, so $\alpha_j/\alpha_i = 1$ iff $i = j$, and the multiplicity is exactly n. □

Remark 7.6 (Scope of the probe)*.* Cyclotomic contact decides coherence *within shells*. Cross-shell coherence is equivalent to $\nu = \mu/\overline{\mu}$ being a root of unity for $\mu = \alpha/\beta$ (note $t(\nu) = 2\,t_{\mathrm{rel}}(\alpha, \beta)$, so the criterion decides rationality exactly while reading the order only up to its 2-part); this is again a torsion question about an algebraic number and is decidable, but it is not needed below: for the ledger's admissible objects the angles are available in closed form and the full type is settled symbolically, while for a Salem number the shells are $\{\tau\}$, $\{1/\tau\}$ and the unit circle, so by Corollary 4.6 the contact scan is a complete decision for the only undetermined block. The shell signature is a strictly coarser invariant than the full type: x^4+5x^2+5 and the catalog seed $K = x^4+5x^2-5$ have identical contact signatures $\{\Phi_1^4, \Phi_2^4\}$ (ledger entries E, F) while their full relational groups are $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$; and the probe is gauge-blind by construction — x^3-2 (absolute $\mathbb{Z}/3\mathbb{Z}$) and x^3+2 (absolute $\mathbb{Z}/6\mathbb{Z}$) return identical signatures $\{\Phi_1^3, \Phi_3^3\}$, both reading the common relational group $\mathbb{Z}/3\mathbb{Z}$ (ledger entries A, B). [forced]

7.2 The two keystone occupants are relationally inert

Definition 7.7 (Relationally inert)**.** A conjugation-closed object is *relationally inert* if every coherence class is either contained in the real axis or a singleton — equivalently, if the only coherent pairs of distinct conjugates are pairs of real conjugates. (Real conjugates are always mutually coherent, their angles lying in $\{0, \frac{1}{2}\}$; inertness says no other coherence exists.)

Theorem 7.8 (Inertness of β_4 and Lehmer's number)**.** *Let $\beta_4 = x^4-x^3-x^2-x+1$ (the minimal degree-4 Salem number, $\mathrm{M} = 1.72208\ldots$) and let L be Lehmer's polynomial $x^{10}+x^9-x^7-x^6-x^5-x^4-x^3+x+1$ ($\mathrm{M} = 1.17628\ldots$). The complete contact scans return*

$$\mathrm{Rat}_{\beta_4}:\ \{\Phi_1^4\} \qquad \textit{and} \qquad \mathrm{Rat}_L:\ \{\Phi_1^{10}\}$$

and nothing else (complete scan bounds $M \le 512 = 2\cdot 16^2$ and $M \le 20000 = 2\cdot 100^2$: Lemma 7.4 applied to $\deg \mathrm{Rat}_{\beta_4} = 16$ and $\deg \mathrm{Rat}_L = 100$). Combined with Corollary 4.6, the coherence types are: rational block $\{\tau, 1/\tau\}$ with trivial class group, plus $2s$ mutually incoherent circle singletons ($s = 1$ for β_4; $s = 4$ for L). In words: both objects are relationally inert (Definition 7.7); the only rational phase relations they possess are the tautological diagonal and the positivity-forced real pair $(\tau, 1/\tau)$. [forced] *per instance (ledger entries G, H)*

Theorem 7.9 (Extension across degrees 4–10)**.** *Let $S_6 = x^6-x^4-x^3-x^2+1$ and $S_8 = x^8-x^5-x^4-x^3+1$. All four polynomials β_4, S_6, S_8, L are Salem, certified exactly: each is reciprocal and irreducible, and its trace polynomial — the $T \in \mathbb{Z}[z]$ with $p(x) = x^{\deg p/2}\,T(x+1/x)$ — has, by Sturm count, exactly one real root in $(2,\infty)$, none in $(-\infty,-2]$, and the remaining $\deg p/2 - 1$ in $(-2,2)$ (ledger entry O). The complete contact scans of Rat_{S_6} and Rat_{S_8} return $\{\Phi_1^6\}$ and $\{\Phi_1^8\}$ respectively (degrees 36 and 64; bounds 2592 and 8192), so both are relationally inert (Definition 7.7; ledger entry P).* [forced] *per instance*

Theorem 7.10 (Exhaustive degree-12 census)**.** *Consider all $3^6 = 729$ monic reciprocal polynomials $x^{12}+c_1x^{11}+\cdots+c_6x^6+\cdots+c_1x+1$ with $c_1,\ldots,c_6 \in \{-1,0,1\}$. The sign twist $p(x) \mapsto p(-x)$ preserves the family and fixes Rat_p (a global sign cancels in every ratio), so the family splits into twist-classes counted by Burnside's lemma: the involution fixes exactly the $3^3 = 27$ coefficient vectors with vanishing odd coefficients, giving $(729+27)/2 = 378$ orbits — an internal count check reproduced by the enumeration. Exactly 37 twist-classes are Salem, each certified as in Theorem 7.9; the complete contact scan of every one ($\deg \mathrm{Rat}_p = 144$, bound $41472 = 2\cdot 144^2$) returns $\{\Phi_1^{12}\}$:* every certified Salem number in the family is relationally inert *(ledger entry T).* [forced] *per instance; the family statement is a complete finite verification,* [computed]

Remark 7.11 (Scope of the census). The $\{-1, 0, 1\}$ family was chosen for exhaustive tractability, not representativeness; with Corollary 7.14 below in hand the census is corroboration rather than evidence. Five further degree-12 Salem instances with a coefficient outside $\{-1, 0, 1\}$ were spot-checked and are likewise inert (ledger entry V). Each of the 341 non-Salem twist-classes carries an exact rejection certificate: 39 vanish at $x = \pm 1$, 257 fail the trace-polynomial Sturm pattern $(1, 0, 5)$, and 45 are reducible (none fail only trace-polynomial irreducibility); the tally sums to 378 (ledger entry W).

Corollary 7.12 (Sharpened exclusion)**.** *For these two instances, [1, Lem. 6.1] strengthens: not only does no finite absolute charge group exist — no nontrivial rational phase relation exists at all among the conjugates off the real axis. The would-be occupants of the odd-charge gap $(\varphi, 2)$ studied in [1, Thm. 6.2] are, at least at the two keystones, maximally far from carrying phase structure in either the absolute or the relational sense. Inertness is a positive finding, not the absence of one: each scan excludes* every *candidate torsion relation among the n^2 ratios — an infinite family of cyclotomic hypotheses — in a single finite certificate, and it is exactly this that upgrades [1, Lem. 6.1] from "no finite charge group" to "no rational phase structure whatsoever off the real axis" for these witnesses. Every Salem number is in fact relationally inert (Corollary 7.14 below); the per-instance decidability of Lemmas 7.3–7.4 remains the general instrument beyond the pinned-modulus case.*

7.3 A general theorem: modulus pinning

Theorem 7.13 (Modulus pinning)**.** *Let $p \in \mathbb{Z}[x]$ be irreducible with a root α_0 whose modulus is attained by no other root of p. Then no two distinct roots of p differ by a root-of-unity factor. More generally, if $q \in \mathbb{Z}[x]$ is irreducible, $q \neq p$, and no root of q has modulus $|\alpha_0|$, then no root of p and root of q differ by a root-of-unity factor.* [forced]

Proof. Suppose $\alpha \neq \beta$ are roots of p with $\alpha/\beta = \zeta$ a root of unity. Let G be the Galois group of the splitting field of p; by irreducibility G acts transitively on the roots, so some $\sigma \in G$ has $\sigma\alpha = \alpha_0$. Then $\sigma\alpha/\sigma\beta = \sigma(\zeta)$ is again a root of unity, so $|\sigma\beta| = |\alpha_0|$; since $\sigma\beta$ is a root of p and α_0 is the only root of that modulus, $\sigma\beta = \alpha_0 = \sigma\alpha$, whence $\alpha = \beta$ — a contradiction. For the mixed statement, work in a Galois extension containing both splitting fields: the restriction map onto Gal of the splitting field of p is surjective, so again some σ carries the p-root of the pair to α_0, and the image of the q-root would be a root of q of modulus $|\alpha_0|$, which does not exist. □

Corollary 7.14 (Every Salem number is relationally inert)**.** *Every Salem number is relationally inert; open problem P1 of the earlier revisions of this note is resolved in the negative, and Theorems 7.8, 7.9, 7.10 become corroborations of a general fact.* [forced]

Proof. The modulus τ is attained only by τ, so Theorem 7.13 applies. A non-singleton class not contained in the real axis would contain either two distinct circle conjugates — coherent iff their ratio is a root of unity (Lemma 7.3), excluded — or a real and a circle conjugate λ: coherence would force $t(\lambda) \in \mathbb{Q}/\mathbb{Z}$, making $\lambda/\overline{\lambda}$ (unimodular, rational angle) a root of unity, again excluded. So every class is real-contained or a singleton, with real block $\{\tau, 1/\tau\}$ of trivial class group. □

Corollary 7.15 (No two Salem numbers circle-lock)**.** *No two distinct Salem numbers are circle-locked; indeed no root of one and root of another differ by a root-of-unity factor. Open problem P2 is resolved in the negative; Theorem 8.2 and Remark 8.4 are corroborations.* [forced]

Proof. Apply the mixed statement with $\alpha_0 = \tau_p$: a root of q of modulus τ_p would force $\tau_p \in \{\tau_q, 1/\tau_q, 1\}$, i.e. $\tau_p = \tau_q$, hence the same minimal polynomial, $p = q$. $\square$

Corollary 7.16 (Pisot numbers)**.** *No two distinct conjugates of a Pisot number differ by a root-of-unity factor, and no non-real conjugate is coherent with a real one. (The full type of a Pisot number can still carry mirrored cross-shell classes; deciding those is the ν-criterion of Remark 7.6.)* [forced]

Proof. The dominant root pins its modulus; the real–non-real piece follows as in Corollary 7.14: coherence with a real conjugate would make $t(\lambda)$ rational and $\lambda/\overline{\lambda}$ a torsion ratio. $\square$

Example 7.17 (A Pisot instance, and the ν-criterion executed)**.** (i) The smallest Pisot number, the plastic number θ_0 (the real root of $x^3 - x - 1$), is relationally inert: its shells are $\{\theta_0\}$ and the conjugate pair at modulus $\theta_0^{-1/2}$ (the root product is $-p(0) = 1$), pinning applies, and the complete contact scan returns $\{\Phi_1^3\}$ (ledger entry W); for a cubic there is no cross-shell residue, so Corollary 7.16 closes the full type. (ii) The ν-criterion of Remark 7.6, executed symbolically on the group-drop quartic $x^4 + 5x^2 + 5$: the cross-shell pair (ia, ib) has $\mu = a/b \in \mathbb{R}_{>0}$, so $\nu = \mu/\overline{\mu} = 1$ exactly — torsion of order 1, hence coherent with $t_{\mathrm{rel}} = 0$, merging the two shells into the single class of Example 3.3. (iii) On β_4's cross-shell pair (τ, λ): $\nu = \overline{\lambda}/\lambda = \lambda^{-2}$ is itself a root of Rat_{β_4}, so its torsion status is decided by the very scan of ledger entry G — no contact, so the pair is incoherent, as pinning independently forces. [forced]

Example 7.18 (Sharpness, and an irreducible non-inert witness)**.** The pinning hypothesis is necessary: $x^4 - 2$, with all four roots on one shell, is irreducible with torsion ratios ζ_4^j (ledger entry C). And irreducible *inadmissible* non-inert objects exist: for $p(x) = q(x^k)$ with q irreducible carrying a non-real root at irrational angle, each fiber of k-th roots is a size-k coherence class with class group $\mathbb{Z}/k\mathbb{Z}$ at an irrational offset. Concretely, $p = x^4 + x^2 + 2 = q(x^2)$ with $q = x^2 + x + 2$: p is irreducible; q's angle is irrational (complete scan of Rat_q: $\{\Phi_1^2\}$, ledger entry U), so p is inadmissible; and Rat_p has signature $\{\Phi_1^4, \Phi_2^4\}$ — the fibers $\pm\sqrt{\rho}$, $\pm\sqrt{\overline{\rho}}$ are non-real size-2 classes, so p is not inert; $\mathrm{M}(p) = 2$. Salem polynomials can never arise this way: $q(x^k)$ with $k \geq 2$ would duplicate the maximal modulus — pinning and the twisted-shell mechanism exclude each other. [forced]

Remark 7.19 (Degeneracy, the Dubickas program, and attribution). In the language of linear recurrence sequences, distinct characteristic roots with a root-of-unity ratio are precisely the *degenerate* case (see [13]). A targeted registry search conducted for this revision (Crossref and the arXiv API, 2 July 2026; terms: *quotients of conjugate algebraic numbers root of unity*; *roots of unity as quotients of two roots of a polynomial*; *multiplicative relations conjugates Salem*; *degenerate recurrence dominant root*; MathSciNet and zbMATH were not available to us) shows the torsion-quotient question to be an established program of Dubickas: quantitative bounds on the least T such that no quotient of two distinct T-th powers of roots of an arbitrary $f \in K[x]$ is a root of unity [14] — a framing that encompasses pairs across irreducible factors and is motivated there by the (non)degeneracy of recurrences — together with [15, 16] on quotients of two conjugates. Only titles, registry metadata, and abstracts (where the registry carries them) were consulted, per the project's provenance discipline; to the extent the pinned-modulus criterion of Theorem 7.13 appears in those works, priority is theirs and the theorem should be read as a self-contained account. What this note claims as its own is the packaging: the pinned-modulus formulation as used here with its mixed two-polynomial form, the coherence-type invariant and its shell-complete decision procedure, the census infrastructure, and the Salem/Pisot corollaries in relational form — corroborated on forty-six instances before the general argument was written down. The multiplicative rigidity of Corollary 7.14 also sits in instructive contrast with the *additive* theory: Dubickas–Jankauskas [17] show nontrivial linear relations among Salem conjugates do exist (from degree 8, descending to relations among conjugates of $\tau + 1/\tau$ — the same trace substitution that certifies our instances), whereas torsion multiplicative relations never do.

8 Cross-object locking

Rigidity kills within-object novelty; it says nothing about relations *between* different oscillator systems. By Proposition 5.3, two objects can each be relationally inert yet be locked — a charge that lives purely in the relationship, with no per-object counterpart. For Salem numbers the interesting case is circle-locking, and it is decidable:

Definition 8.1 (Mixed ratio object)**.** For monic $p, q \in \mathbb{Z}[x]$ with nonzero constant terms, the primitive part of $\mathrm{Rat}_{p,q}(x) = \mathrm{Res}_y\big(q(y),\, p(xy)\big)$ has root multiset $\{\alpha/\beta :\ p(\alpha) = 0,\ q(\beta) = 0\}$, of degree $(\deg p)(\deg q)$.

For two Salem numbers the unimodular cross-ratios are exactly the circle-by-circle ones (the real conjugates of distinct Salem numbers never share a modulus: $\tau_p = \tau_q$ is excluded by distinctness and $\tau_p = 1/\tau_q$ by $\tau > 1$), so the contact scan on $\mathrm{Rat}_{p,q}$ is a complete decision for circle-locking; contact with Φ_1 would mean a shared root, excluded when $\gcd(p, q) = 1$. (Corollary 7.15 now settles circle-locking for the entire Salem class; the scans below stand as the original exact corroboration.)

Theorem 8.2 (β_4 and Lehmer's number are not circle-locked)**.** $\gcd(\beta_4, L) = 1$*, and the complete contact scan of the degree-*40 *mixed ratio object* Rat_{L,β_4} *(complete bound* $M \leq 3200 = 2 \cdot 40^2$*: Lemma 7.4 applied to* $\deg \mathrm{Rat}_{L,\beta_4} = 40$*) returns* no *cyclotomic contact whatsoever. Hence no circle conjugate of Lehmer's number differs from a circle conjugate of* β_4 *by a rational angle: the two keystone occupants of* $(\varphi, 2)$ *are mutually as incoherent as each is internally.* [forced] *per instance (ledger entry I)*

Remark 8.3. The trivial real locking $\tau_{\beta_4} \approx \tau_L$ (both at angle 0) is of course present; it is the rational-block locking that Proposition 5.3 discounts. What Theorem 8.2 rules out is the informative kind.

Remark 8.4 (All pairs among the certified instances)*.* The same verdict holds for every pair among the four instances of Theorem 7.9: the five remaining mixed ratio objects ($\beta_4 \times S_6$, $\beta_4 \times S_8$, $S_6 \times S_8$, $L \times S_6$, $L \times S_8$; degrees $24, 32, 48, 60, 80$, complete bounds 1152–12800) carry no cyclotomic contact (ledger entry Q). No two of the four certified Salem numbers are circle-locked. [forced] per instance

Example 8.5 (A non-inert object, computed: $\beta_4 \otimes \beta_4$)**.** Write β_4's conjugates as $\tau, 1/\tau, \lambda, \overline{\lambda}$, with λ on the unit circle at irrational angle θ. The 16 eigenvalues of $\beta_4 \otimes \beta_4$ sort into the coherence type: a rational block of six positive reals $\{1, 1, 1, 1, \tau^2, \tau^{-2}\}$ at offset 0; a mirrored pair of size-4 classes at offsets $\pm\theta$, namely $\{\tau\lambda, \tau\lambda, \tau^{-1}\lambda, \tau^{-1}\lambda\}$ and its conjugate — two distinct values, each of multiplicity two, sharing a single angle, so the class group is trivial while the class is non-real of size 4; and a mirrored singleton pair $\{\lambda^2\}, \{\overline{\lambda}^2\}$ at offsets $\pm 2\theta$. By Definition 7.7 the object is *not* inert: the offset-cancellation mechanism of Proposition 5.2(iii) manufactures both a rational block and non-real coherence from an inert factor. Engine corroboration (ledger entry S): the characteristic polynomial F of the 16×16 Kronecker square $C_{\beta_4} \otimes C_{\beta_4}$ has $(x-1)$-multiplicity exactly 4, $\deg \gcd(F, F') = 7$ (exactly the multiplicity pattern $1^4 \cdot$ (four doubles)), and exactly 3 distinct real roots, all positive — Sturm counts, no floats. [forced]

9 Methodology

Exact arithmetic, strengthened. The companion note decided angle questions at 25–35 decimal digits with a tolerance discipline. The relational questions of this note are *torsion* questions, and the entire verification path here is integer and rational polynomial arithmetic: resultants over $\mathbb{Z}[x]$, primitive parts, and exact trial division by cyclotomic polynomials. No floating-point number appears anywhere — not in a decision, not in a displayed value. The completeness certificate of Lemma 7.4 replaces any notion of numerical tolerance.

Engine. `sympy` 1.14.0 on Python 3.12.3. The consolidation ledger of Appendix A was produced by the scripts `relational_charge_probe.py` and `supplement_for_paper.py` (Appendix B lists the probe core), each rerun in full immediately prior to writing; every numerical claim in the text is grounded in a ledger entry. SHA-256 digests of the deliverables (source, PDF, scripts) accompany the artifacts in `SHA256SUMS`.

Second engine. Following review, the complete contact signatures of ledger entries A–I, together with rows P, Q, U, and V, were recomputed independently in PARI/GP 2.15.4 via `polresultant`, integer factorization, and `poliscyclo`; all twenty-three signatures agree with the `sympy` computations (ledger entry M). The two stacks share no computer-algebra code: `sympy`'s resultants and factorizations are pure Python, PARI's are its own C library.

Two-route corroboration. For β_4 the ratio object was computed independently by the resultant of Definition 7.2 and as the characteristic polynomial of the 16×16 integer matrix $C_{\beta_4} \otimes C_{\beta_4}^{-1}$; the results agree (ledger entry G).

A harness incident, recorded. The first assertion written for the control q_2 predicted Φ_2-multiplicity 2 and failed; the engine was right and the assertion wrong — ratios are *ordered* pairs, so each antipodal pair contributes twice. Per the project discipline that load-bearing corrections are pinned rather than papered over, the incident is recorded here and the corrected expectation (Φ_2^4) is ledger entry D.

Limits of self-verification. The cross-engine run was performed by the same operator as the primary computation; the stacks are disjoint, the operator is not. The polynomial inputs are generated programmatically from coefficient vectors, confining operator-dependent transcription to the enumeration logic — itself validated by the Burnside count of Theorem 7.10 matching the enumeration (378 twist-classes). Independent replication by external researchers is welcomed: the scripts, inputs, expected signatures, and digests are provided for exactly this purpose (`relational_charge_probe.py`, `supplement_for_paper.py`, `supplement_round3.py`, `census_deg12.py`, `supplement_round6.py`, `supplement_round7.py`, `cross_check.gp`, `cross_round6.gp`, `SHA256SUMS`); the project repository is DOI-archived at `10.5281/zenodo.21121863`; the scripts above now ship in that deposit (the complete artifact set as of v1.0.2, version DOI `10.5281/zenodo.21123586`, in the `relational-charge-verification/` package alongside this manuscript), enabling verification by a distinct operator. Concretely (expected outputs are the signatures of Appendix A; any deviation is a finding):

- `python3 relational_charge_probe.py` — entries A–H under assertions (e.g. $\{\Phi_1^{10}\}$ for Lehmer; about one second);
- `python3 supplement_round3.py` — entries O–S;
- `python3 supplement_round6.py` — entries U–V;
- `python3 supplement_round7.py` — entry W (plastic number; census rejection tally; symbolic ν-criterion);

- `python3 census_deg12.py` — the census of Theorem 7.10 (about two minutes);
- `gp -q cross_check.gp` and `gp -q cross_round6.gp` — the twenty-three cross-engine signatures.

10 Scope, limitations, and open problems

Result	Statement	Status
Cyclicity (Prop. 3.4)	class/relational groups cyclic	[forced]
Gauge (Prop. 3.5)	absolute = relational + ray	[forced]
Rigidity (Thm. 4.1)	relational adm. $\Rightarrow$ absolute; $n \in \{m, 2m\}$	[forced]
Structure (Thm. 4.3)	rational block + mirrored offset classes	[forced]
Odd anchors full (Lem. 4.4)	n odd $\Rightarrow \Delta = \mathbb{Z}/n\mathbb{Z}$	[forced]
Descent (§5)	lcm law, CRT, safe $\Leftrightarrow$ coprime	[forced]
Internalization (Prop. 5.2iii)	$\otimes$ creates rational blocks by offset cancellation	[forced]
Golden relation (Lem. 6.1)	$t_{\mathrm{rel}}(\varphi, \varphi') = \frac{1}{2}$	[forced]
Twist (Lem. 6.3)	anchor sectors exchanged at odd m, M fixed	[forced]
Parity floor (Thm. 6.5)	$\mu_{\mathrm{rel}}(\text{even}) = \varphi$ attained	[forced]
	$\mu_{\mathrm{rel}}(\text{odd}) = \mu(\text{odd})$; value 2	[forced] at 3; [computed]/[plausible] general
	lower bound φ	conditional on (EG), Rem. 2.3
Inertness (Thm. 7.8)	β_4, Lehmer relationally inert	[forced] per instance
Extension (Thm. 7.9)	S_6, S_8 certified Salem; inert	[forced] per instance
Pinning (Thm. 7.13)	pinned modulus forbids torsion ratios	[forced]
Salem inertness (Cor. 7.14)	every Salem number inert; P1 closed	[forced]
No locking (Cor. 7.15)	no two Salem numbers circle-lock; P2 closed	[forced]
Not locked (Thm. 8.2, Rem. 8.4)	six pairwise scans	corroboration
Census (Thm. 7.10)	37/37 degree-12 family Salems inert	corroboration; [computed] family
Cross-engine (App. A, entry M)	PARI/GP reproduces signatures A–I, P, Q	[computed]

Open problems.

(P1) **Internal Salem coherence — resolved in the negative.** Corollary 7.14: every Salem number is relationally inert. The forty-six certified instance decisions of this note — the four canonical test cases of degrees 4–10, the exhaustive degree-12 family, and the five larger-coefficient spot checks (ledger entries G, H, P, T, V) — stand as exact corroboration, obtained *before* the theorem. The successor problem is the characterization: **(P1′)** which irreducible integer polynomials carry nontrivial coherence? A torsion ratio ζ_M forces $\mathbb{Q}(\zeta_M)$ into the splitting field; a pinned modulus forbids torsion (Theorem 7.13); twisted shells $q(x^k)$ suffice (Example 7.18); and composite objects manufacture coherence freely (Example 8.5).

In recurrence language: characterize degenerate irreducibles beyond the pinned/twisted-shell dichotomy (cf. [13]; the quantitative side of exactly this question is developed in [14]), and, beyond torsion, classify the cross-shell mirrored classes the ν-criterion governs. [open]

(P2) **Cross-Salem locking — resolved in the negative.** Corollary 7.15: the circle angles of distinct Salem numbers, viewed in $(\mathbb{R}/\mathbb{Z})/\mathbb{Q}$, never collide. Theorem 8.2 and Remark 8.4 (six pairs) are exact corroboration. [forced]

(P3) **The odd floor.** Prove $\mu_{\mathrm{rel}}(m) = 2$ for all odd m. By Lemma 6.3 this is *exactly* the companion's principal open problem [1, §8] — the relational refinement adds no new burden, since the new anchor sector reduces by the twist. [open]

(P4) **Is parity the whole coupling?** Does μ_{rel} depend on Δ through anything besides the single bit $2 \mid m$? The values known and conjectured (Theorem 6.5) suggest not. [open]

Caveat on the larger programme. As in [1], nothing here touches Lehmer's problem itself or the substrate's deeper conjectures. What this note establishes is internal to the phase structure: the absolute charge was a gauge-fixed presentation of a relational invariant; integrality is what fixes the gauge; the parity coupling survives the deletion of the reference ray unchanged; and on the sector the absolute theory discards, the relational theory is nontrivial, finite, decidable — and, at the two keystones of the gap $(\varphi, 2)$, maximally empty.

A Consolidation ledger

The following were regenerated in exact arithmetic immediately prior to writing (engine of §9); each grounds a claim in the text. "Contacts" means the complete cyclotomic-contact signature of the stated ratio object (Lemmas 7.3–7.4), written $\{\Phi_M^{\mathrm{mult}}\}$. Backing scripts:

`relational_charge_probe.py` and `supplement_for_paper.py`.

	object	exact result
A	x^3-2	contacts $\{\Phi_1^3, \Phi_3^3\}$; $\Delta = \mathbb{Z}/3\mathbb{Z}$, anchor $n=3$, $\mathrm{M}=2$.
B	x^3+2	contacts $\{\Phi_1^3, \Phi_3^3\}$, identical to A: probe is gauge-blind ($\Delta = \mathbb{Z}/3\mathbb{Z}$; anchor $n=6$); twist $T(x^3+2) = x^3-2$.
C	x^4-2	contacts $\{\Phi_1^4, \Phi_2^4, \Phi_4^4\}$: relational recovery of $\mathbb{Z}/4\mathbb{Z}$ with no reference ray.
D	$q_2 = x^4+x^2-1$	contacts $\{\Phi_1^4, \Phi_2^4\}$ (ordered-pair multiplicities; see §9); full $\Delta = \mathbb{Z}/4\mathbb{Z}$ settled symbolically; $\mathrm{M} = \varphi$.
E	x^4+5x^2+5	irreducible (Eisenstein at 5); x^2-roots $\frac{-5\pm\sqrt{5}}{2}$ both negative, so all roots purely imaginary; anchor $n=4$, $\Delta = \mathbb{Z}/2\mathbb{Z}$ (group drop), $\mathrm{M}=5$; contacts $\{\Phi_1^4, \Phi_2^4\}$.
F	$K = x^4+5x^2-5$	contacts $\{\Phi_1^4, \Phi_2^4\}$, identical to E, while the full type differs: angles $\{0, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}\}$, so $\Delta = \mathbb{Z}/4\mathbb{Z}$ — the shell signature is strictly coarser than the type.
G	$\beta_4 = x^4-x^3-x^2-x+1$	contacts $\{\Phi_1^4\}$ only (complete scan $M \le 512$): circle block inert. Two-route: resultant route = characteristic polynomial of $C_{\beta_4} \otimes C_{\beta_4}^{-1}$.
H	Lehmer L, degree 10	$\deg \mathrm{Rat}_L = 100$; contacts $\{\Phi_1^{10}\}$ only (complete scan $M \le 20000$, 1.1 s): all 8 circle conjugates mutually inert.
I	$\beta_4 \times L$ mixed	$\gcd(\beta_4, L) = 1$; $\deg \mathrm{Rat}_{L,\beta_4} = 40$; contacts $= \varnothing$ (complete scan $M \le 3200$): not circle-locked.
J	golden pair	$\varphi/\varphi' = -\varphi^2$: negative real, so $t_{\mathrm{rel}}(\varphi, \varphi') = \frac{1}{2}$.
K	y^2+y-1	roots $1/\varphi > 0$ and $-\varphi < 0$: the real $\pm$ pair behind Lemma 6.2, all k.
L	coset arithmetic	e odd, m odd $\Rightarrow (e+m)/2m \in \frac{1}{m}\mathbb{Z}/\mathbb{Z}$; engine corroboration of Lemma 6.3's one-line step, $m \in \{3,5,7,9\}$, all odd $e < 2m$.
M	cross-engine replication	contact signatures of entries A–I, P, Q, U, and V recomputed in PARI/GP 2.15.4 (`polresultant` + `poliscyclo`); all twenty-three identical to the `sympy` signatures.
N	totient bound	$2\,\phi(M)^2 \ge M$ verified by exact integer sieve for all $1 \le M \le 2 \cdot 10^5$; unique tight case $M=2$ (Lemma 7.4).
O	Salem certification	β_4, S_6, S_8, L: each reciprocal and irreducible; trace-polynomial Sturm counts $(1, 0, \deg p/2 - 1)$ in $(2,\infty)$, $(-\infty,-2]$, $(-2,2)$ — all four Salem, exactly (Theorem 7.9).
P	S_6, S_8 inertness	Rat_{S_6} (deg 36, bound 2592): contacts $\{\Phi_1^6\}$; Rat_{S_8} (deg 64, bound 8192): contacts $\{\Phi_1^8\}$.
Q	pairwise locking batch	the five remaining pairs among $\{\beta_4, S_6, S_8, L\}$: mixed ratio objects of degrees $24, 32, 48, 60, 80$ (bounds 1152–12800); contacts $= \varnothing$ in every case.
R	quadratic identities	$(1+\sqrt{5})/2 = \varphi$ and $(3+\sqrt{5})/2 = \varphi^2$ (Appendix D).
S	$\beta_4 \otimes \beta_4$ (non-inert witness)	charpoly F of the 16×16 Kronecker square: $(x-1)$-multiplicity $= 4$; $\deg \gcd(F, F') = 7$ (pattern 1^4·four doubles); 3 distinct real roots, all positive (Sturm). Structure forced symbolically; single-engine run (Example 8.5).
T	degree-12 census	729 reciprocal $\{-1,0,1\}$-polynomials; 378 twist-classes (27 twist-fixed; $378 = (729+27)/2$); 37 certified Salem; all 37 complete scans (deg 144, bound 41472) return $\{\Phi_1^{12}\}$. Runtime 93 s (`census_deg12.py`); cross-engine for the four canonical instances via entry M.
U	twisted-shell witness	$q = x^2+x+2$: Rat_q contacts $\{\Phi_1^2\}$ (angle irrational, complete); $p = x^4+x^2+2 = q(x^2)$: irreducible, Rat_p contacts $\{\Phi_1^4, \Phi_2^4\}$, $\mathrm{M}=2$ — non-inert (Example 7.18).
V	larger-coefficient spot checks	first five degree-12 Salem twist-classes with a coefficient outside $\{-1,0,1\}$ (lexicographic in $\{-2,\dots,2\}^6$): all five scans return $\{\Phi_1^{12}\}$ (Remark 7.11).
W	plastic number; tally	x^3-x-1: irreducible, one real root > 1 (Sturm), pair modulus $\theta_0^{-1/2}$; contacts $\{\Phi_1^3\}$ — inert (Example 7.17). Census rejection tally re-derived: 39/257/45/0/37, sum 378. Single-engine (`supplement_round7.py`).

B The probe core

The complete decision procedure of Lemmas 7.3–7.4, as executed. Everything is integer polynomial arithmetic; `x, y` are `sympy` symbols.

```
def ratio_poly(p_expr):
    R = sp.resultant(p_expr.subs(x, y), sp.expand(p_expr.subs(x, x*y)), y)
    return sp.Poly(sp.expand(R), x).primitive()[1]

def totients_upto(N):
    phi = list(range(N + 1))
    for i in range(2, N + 1):
        if phi[i] == i:                      # i prime
            for j in range(i, N + 1, i):
                phi[j] -= phi[j] // i
    return phi

def cyclotomic_contacts(P):
    d = P.degree()
    N = 2 * d * d                # complete: Phi_M | P forces M <= 2 d^2
    phi = totients_upto(N)
    hits = {}
    for m in range(1, N + 1):
        if phi[m] > d:
            continue
        cm = sp.Poly(sp.cyclotomic_poly(m, x), x)
        if sp.rem(P, cm).is_zero:  # Phi_m irreducible: divides, or no contact
            mult, T = 0, P
            while True:
                q, r = sp.div(T, cm)
                if r.is_zero:
                    mult, T = mult + 1, q
                else:
                    break
            hits[m] = mult
    return hits
```

The PARI/GP cross-check (ledger entries M, P, Q) is the following independent implementation; its expected outputs are exactly the contact signatures of Appendix A.

```
contacts(R) = {
  my(F = factor(R), v = List());
  for(i = 1, matsize(F)[1],
    if(type(F[i,1]) == "t_POL" && poldegree(F[i,1]) > 0,
      my(n = poliscyclo(F[i,1]));
      if(n, listput(v, [n, F[i,2]]))));
  vecsort(Vec(v))
}
```

```
rat(p)      = polresultant(subst(p, x, 'y), subst(p, x, x*'y), 'y);
mixed(p, q) = polresultant(subst(q, x, 'y), subst(p, x, x*'y), 'y);
```

The mixed probe replaces the first argument of the resultant by the second polynomial: $\mathrm{Rat}_{p,q} = \mathrm{Res}_y\big(q(y),\, p(xy)\big)$.

C Inherited statements from the companion note

Reference [1] is a technical note of the same project (the `math-research-pipelines` corpus), distributed alongside this one and archived with the repository at DOI `10.5281/zenodo.21121863`. So that the present note can be read, and its dependence structure audited, without [1] in hand, the table below lists every statement imported from it, the status *as tagged there*, and where it enters this note. Nothing is silently promoted; items this note re-derives are marked.

[1] item	statement	status in [1]	used here
Def. 2.4	absolute admissibility; charge group $\mathbb{Z}/n\mathbb{Z}$; charge χ_n	definition	Def. 2.2
Thm. 3.1	charge groups are finite cyclic	[forced]	Prop. 3.4 (re-derived)
Thm. 3.2	$x^n - 2$ realizes $\mathbb{Z}/n\mathbb{Z}$, $\mathrm{M} = 2$	[forced]	Thm. 4.1, Thm. 6.5
Thm. 3.3	CRT via Adams operators	[forced]	Prop. 5.1
Thm. 4.2	lcm law under $\otimes$	[forced]	Prop. 5.2 (re-derived)
Thm. 4.3	safe composition $\Leftrightarrow$ coprime	[forced]	Prop. 5.2
Prop. 2.3	emission gap: $\mathrm{M} \in \{1\} \cup [\varphi, \infty)$ for admissible objects	[forced] (quadratics); [computed]/[plausible] (general); named (EG) here	Rem. 2.3, Thm. 6.5
Thm. 5.1	absolute parity floor $\mu(\text{even}) = \varphi$, $\mu(\text{odd}) = 2$	even [forced]; odd [forced] at $n = 3$, [computed]/[plausible] general	Thm. 6.5
Lem. 5.2	π-ray obstruction for the golden seed	[forced]	Lem. 6.1, Cor. 6.7
Lem. 6.1	Salem numbers are charge-inadmissible	[forced]	Cor. 4.6, §7
Thm. 6.2	odd gap $(\varphi, 2)$ = no-Salem closure	[forced] at $n = 3$; [computed]/[plausible] general	Thm. 6.5, §8

D The quadratic case of (EG), self-contained

For completeness we reproduce the elementary argument behind the [forced] portion of (EG): *no charge-admissible object of degree at most* 2 *has Mahler measure in* $(1, \varphi)$.

Degree-1 objects and reducible quadratics have nonzero integer roots, so $\mathrm{M} \in \{1\} \cup \mathbb{Z}_{\geq 2}$. Let $p = x^2 - tx + n$ be irreducible, $t, n \in \mathbb{Z}$, $n \neq 0$.

Complex pair. Then $n = \alpha\overline{\alpha} = |\alpha|^2 \geq 1$. Admissibility puts α on a rational ray; if $n = 1$ then α is unimodular with rational angle, hence a root of unity and $\mathrm{M} = 1$; if $n \geq 2$ then $\mathrm{M} = n \geq 2$.

Real pair, $|n| \geq 2$. Real conjugates are automatically admissible (angles in $\{0, \frac{1}{2}\}$), and not both roots can lie in the closed unit disk (their product has modulus ≥ 2). If both lie outside, $\mathrm{M} = |n| \geq 2$; if one lies strictly inside, $\mathrm{M} = |n|/|\text{inner root}| > |n| \geq 2$.

Real units, $n = -1$. The roots are $\frac{t \pm \sqrt{t^2+4}}{2}$; $t = 0$ gives the reducible $x^2 - 1$, and for $|t| \geq 1$,

$$\mathrm{M} = \frac{|t| + \sqrt{t^2+4}}{2} \;\geq\; \frac{1+\sqrt{5}}{2} = \varphi,$$

with equality exactly at $|t| = 1$ (ledger entry R): both gaps $|t| - 1 \geq 0$ and $\sqrt{t^2+4} - \sqrt{5} \geq 0$ are nonnegative.

Real units, $n = +1$. Real roots force $t^2 > 4$, i.e. $|t| \geq 3$, and $\mathrm{M} = \frac{|t|+\sqrt{t^2-4}}{2} \geq \frac{3+\sqrt{5}}{2} = \varphi^2 > \varphi$ (ledger entry R); $|t| \leq 2$ gives only cyclotomic or reducible cases.

Hence every quadratic value of M lies in $\{1\} \cup [\varphi, \infty)$, with φ attained by $x^2 - x - 1$. Beyond degree 2, (EG) remains [computed]/[plausible] as recorded in Remark 2.3; the structural reason the gap is expected to survive — the would-be occupants of $(1, \varphi)$ near rational-angle lattices are Salem/Pisot numbers with irrational angles, which are inadmissible — is [1, Lem. 6.1, Thm. 6.2].

References

[1] *The Charge–Measure Coupling on a Spectral Semiring: Cyclicity, Composition, and a Parity-Graded Mahler Floor*, math-research-pipelines technical note, 30 June 2026. Archived within the project repository deposit at DOI `10.5281/zenodo.21121863` (v1.0.0, as `papers/charge_measure_coupling.tex`); served at `https://echo-s-studios.github.io/math-research-pipelines/`. Appendix C lists every statement imported from it.

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Citation provenance: reference [1] is the companion note of this project (DOI `10.5281/zenodo.21121863`). The bibliographic metadata of [2]–[8] was machine-checked against the Crossref registry on 1 July 2026. Confirmed in full: [3] (DOI `10.2307/1968172`), [5] (`10.1112/jlms/s1-37.1.341`), [6] (`10.1112/blms/3.2.169`), and [8] (`10.1007/978-3-0348-8632-1`, Birkhäuser, 1992). Confirmed with registry quirks noted: [2] — title, year 1857, and pp. 173–175 confirmed; Crossref registers old Crelle volumes by year, while the DOI `10.1515/crll.1857.53.173` encodes the traditional volume 53 used here. [7] — title, volume 24, and pp. 453–469 confirmed at DOI `10.4153/cmb-1981-069-5`, whose registry issued-year field reads 1980 while the DOI and the standard citation year are 1981, retained here. [4] — title, year 1945, and volume 12 confirmed at DOI `10.1215/s0012-7094-45-01213-0`; the registry record carries no page field, so the range 153–172 stands as supplied. References [9], [10], [11], and [12], added at review, were machine-checked likewise, at DOIs `10.1215/s0012-7094-77-04413-1` (Duke's registry record, as for [4], carries no page field, so 315–328 stands as supplied), `10.4064/aa-34-4-391-401`, `10.1016/0022-314x(86)90094-6`, and `10.1090/s0025-5718-98-01006-0`. Reference [13] was machine-checked at DOI `10.1090/surv/104`. References [14], [15], [16], and [17], added at this revision following the novelty search of Remark 7.19, were machine-checked at DOIs `10.1017/s1446788712000183`, `10.3336/gm.52.2.03` (this record's registry author field is partially malformed; the resolvable author is retained), `10.1134/s0037446621030034`, and `10.5802/jtnb.1116`; abstracts, where present in the registry, were consulted; full texts were not.